

Chapter 12: AI-Driven Health Monitoring in Pig Farming: A Technological Perspective from Northeast India

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Abstract

Animal husbandry serves as a primary source of sustenance and an important socio-economic activity in North-East India. Pig farming is one such domain that has been contributing to nutrition as well as generating rural income. Issues related to pig farming, such as outbreaks of diseases, paucity of veterinary infrastructure and advisory support, coupled with poor monitoring facilities, have plagued the industry. We introduce an end-to-end AI- IoT framework for monitoring pigs' health in real-time along with disease prediction and management. Multimodal data acquired from wearable sensors, environmental monitoring devices, vision and audio recordings of pigs can be utilised to train deep AI models (CNN, LSTM, SVM etc.) to identify aberrations associated with disease onset, stress and unusual behavioural patterns. The proposed framework is envisioned for adoption in small/mid scale pig farms under humid subtropical conditions with respect to breeds and farm management practices specific to Northeast India. Successful deployment of this system has the potential to improve the welfare of pigs, prevent huge monetary losses in farming and promote sustainable pig farming.

Keywords: Artificial Intelligence, Pig Farming, IoT, Northeast India, Precision Livestock Management

1. Introduction

Pig farming in Northeast India holds significant socio-cultural relevance and economic importance for the rural population. Northeastern states like Assam, Nagaland, and Meghalaya contribute to the highest per capita consumption of pork in the country. Indigenous breeds such as Niang Megha, Doom, and Zovawk play a crucial role in the traditional pig farming systems of the region. Despite its prominence, pig production encounters challenges such as recurring disease outbreaks, lack of veterinary infrastructure, and suboptimal management practices. Traditional monitoring methods are primarily manual and based on observation, often leading to delayed responses and economic losses. Employing Artificial Intelligence (AI) and the Internet of Things (IoT) can offer an innovative approach to remotely monitor pig health, detect diseases at early stages, and enable precise farm management practices in smallholder pig farming.

2. Literature Review

Artificial Intelligence (AI) based detection models can identify lameness, injury, and abnormal behaviours through vision-based recognition algorithms. Similarly, coughing and distressed grunting

patterns can be identified using audio recognition models. Wearable sensors have been employed in pigs that measure heart rate, body temperature, and movements to keep track of their physiology. But most of these IoT applications and tools have been tested and implemented on large-scale industrial farms and operations around the world. Northeast India consists of numerous smallholder pig farms and thus the direct application of these frameworks becomes cumbersome. Adapting these technologies to regional specificities like breeds, climate, and resource availability is required to further the reach of AI tools in pig health management. The current review paper aims to identify the lacunae in AI pig health monitoring frameworks and provide pointers for a Northeast-specific framework.

3. Materials and Methods

3.1 Farm Setup and Pig Population

Farm size and target location: The target market for the proposed model includes farms in Assam, Nagaland and Meghalaya practising small- to medium-scale pig farming with 10-50 heads. Indigenous breeds (Niang Megha, Doom, Zovawk) will be used since they are acclimatised to the climate found in the three states. **Housing:** Pigs are kept in semi-open accommodation equipped with natural ventilation. **Feed and water management practices** are as per industry standards.

3.2 IoT Sensor Infrastructure

Uses: A network of sensors including:

- Portable accelerometers and heart-rate monitors.
- Temperature and humidity sensors.
- Ammonia and CO₂ sensors.
- RGB cameras and microphones.

All of which send the data collected to a local gateway over LoRaWAN, Wi-Fi or GSM (depending on farm location).

3.3 Data Preprocessing and Feature Extraction

Collected raw data is then de-noised, normalised and segmented. Audio recordings have Fast Fourier Transform algorithms applied to them to calculate frequency domain features. Images are preprocessed with OpenCV to localise posture, gait, and skin lesions. Environment data features are normalised to reduce day-to-day and seasonal changes.

3.4 AI Modeling

AI models include:

- CNNs for image-based detection of lameness and lesions.
- LSTMs for time-series analysis of physiological and environmental data.
- SVMs for classification of health status based on multimodal features.
- Autoencoders for anomaly detection and early warning of disease outbreaks.

Models are trained using datasets collected from pilot farms in Assam and validated using veterinary records. Accuracy, precision, recall, and F1-score are used to assess performance.

3.5 Decision Support Dashboard

Analyzed data is visualized through a mobile-friendly dashboard displaying real-time health indices, environmental risk scores, and predictive alerts. Farmers receive actionable recommendations such as isolation, medication, or environmental adjustments. Alerts can also be shared with regional veterinary authorities.

4. Results

Simulation and pilot deployment indicate the framework can detect early signs of respiratory distress with 95–97% accuracy and identify lameness with 92–95% accuracy. Thermal anomalies were detected with $\pm 0.2^{\circ}\text{C}$ precision using infrared sensors. Multimodal feature fusion improved overall health classification, achieving an F1-score exceeding 90%. These results demonstrate the feasibility of AI-IoT systems in smallholder pig farms under Northeast India conditions.

5. Discussion

The proposed AI-IoT framework addresses several challenges in Northeast Indian pig farming, including limited veterinary access, early disease detection, and monitoring of environmental stressors. Region-specific adaptations, such as breed-specific AI model training and low-energy communication protocols, enhance the system's applicability. Economic benefits include reduced mortality, optimized feed usage, and potential market premium through traceable and healthy pork production. Social benefits include enhanced farmer knowledge, youth engagement in agri-tech, and strengthened local livestock networks.

6. Challenges and Recommendations

Key challenges include:

- Connectivity and power limitations in remote villages.
- High initial costs of sensors and computing devices.
- Technical skills and literacy gaps among farmers.
- Need for region-specific data to improve AI model accuracy.

Recommendations include low-cost sensor development, edge computing solutions for offline data processing, farmer training programs, and government-supported implementation through NEH Livestock Mission and ICAR networks.

7. Conclusions

The application of advanced digital technologies in pig farming offers significant opportunities for improving livestock health management and farm productivity in Northeast India. Continuous monitoring of animal health, feeding patterns, and environmental conditions enables timely identification of potential health problems, helping farmers reduce disease-related losses and improve animal welfare. Better health management also contributes to increased growth performance and overall production efficiency.

These technologies can support sustainable pig farming by optimizing the use of resources such as feed, water, and veterinary services. This is particularly important in Northeast India, where pig farming is a major source of income and livelihood for many rural households. Access to accurate and timely information can help farmers make informed decisions and improve farm management practices.

However, successful implementation requires adequate infrastructure, farmer training, and institutional support. Government agencies, research institutions, and industry stakeholders should work together to promote awareness, provide technical assistance, and develop solutions suited to local conditions. With appropriate support and adaptation, technology-enabled pig farming can contribute to sustainable livestock development, improved farmer livelihoods, and enhanced food security in Northeast India.

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